

OPENCV AI COMPETITION 2026 · AGENTIC VISION

afterimage

A visual inspection agent that remembers. It aligns every photo to the stored baseline of the same asset with OpenCV 5, and a measured number decides what it does next.

PROBLEM

What matters is what changed since the last time anyone looked.

~2,900

modules per MW of solar plant

~15%/yr

lost to PID in affected modules,
partly reversible if caught early

5–20%

of annual energy lost to soiling,
depending on site

Nobody can look at them all, twice. Every prior winner analysed one frame or one session; none of them remember.

SOLUTION

An agent that keeps the memory of each asset and acts on the difference.



Solar panels

hot spots, delamination, cracked glass
— measured



Metal structures

corrosion — demonstration



Concrete

cracks — demonstration

DEMO

DEMO-PANEL-B-61866F · INSPECTED 4 OCT 2026 · 20:53 UTC

A person has to look at this change

5 tool calls, 5 thresholds, 1.69 s. Every step below was decided by a fixed rule, never a guess, and each one shows the number that triggered it.

human_approval

awaiting approval

detector: aliked+lightglue



before

see the full photo



now

Is this real damage?

Compare the same spot before and now. If it is damage, confirm it and it is saved to the panel's history. If it is something else (glare, a shadow), dismiss it and nothing is saved.

Confirm damage

what was it? (optional)

e.g. glare

Dismiss

THE NUMBER THAT DECIDED IT

0.5945

score

0

threshold 0.4

1

score 0.5945 >= 0.4 -> human_approval

How bad the change is, from 0 to 1. Above the limit a person has to look before anything is saved.

A severe change waits for a person

score 0.5945 ≥ 0.4 →

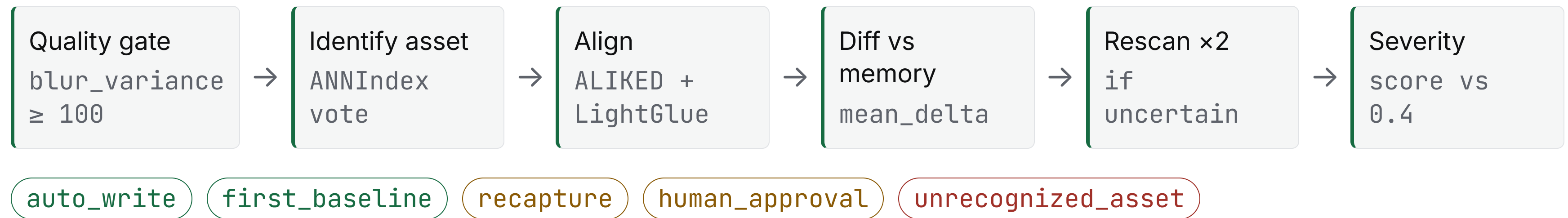
human_approval

The card asks “Is this real damage?” and shows the changed spot zoomed, before and now, beside **Confirm damage** and **Dismiss**. Nothing is written to memory until someone decides.

Five sample sets in /app walk all four answers with no photo of your own.

HOW IT WORKS

Perception reports numbers. One policy file decides.



Six OpenCV 5 tools over MCP, in an arm64 container on AWS Lambda (Graviton). Memory in DynamoDB and S3. Gemini orders the tool calls and phrases the result; it cannot move a threshold or skip a step. Every decision is a hash-chained trace event: {metric, value, threshold, branch}.

RESULTS

29 scenarios, 18 on real photographs.

24 / 29

scenarios pass every assertion

0.8621

branch accuracy

0.8542

defect macro F1

0.7875

mean IoU of the located region

20.4 s

per inspection, p50

\$0.0005

per inspection at list price

The five failures are published with their root cause. CI fails if a headline figure drifts from a fresh run.

JUDGING CRITERIA

30%	Technical execution	OpenCV 5 alignment with ORB fallback, 380 tests, eval gate in CI
20%	Innovation	longitudinal memory per asset; rescans on its own
20%	Real-world impact	solar, metal and concrete inspection
10%	User experience	sample sets, before/now approval, EN/ES, plain/technical, dark
10%	Docs and presentation	technical report, field manual, video
10%	Cloud and responsible ops	Lambda + CloudFormation over OIDC, calibration gate

30%	OpenCV 5 + agent	six OpenCV tools exposed over MCP
25%	Orchestration, autonomy	enforced next tool; the model cannot change a branch
20%	Effectiveness, evaluation	29 reproducible scenarios, failures published
15%	Failure, trace, human	recapture, refusal, approval gate, hash-chained trace
10%	UX, docs, demo	walk all four answers in a minute

What it does not claim.

- **Dataset, not field accuracy.** The figures hold for the committed dataset; plant imagery outside it is the next step.
- **Solar only is measured.** The metal and concrete sets are demonstrations with one synthetic defect each.
- **Five scenarios fail**, each traced to a cause: a global coverage default, the exposure gate firing late, severity ignoring the class, a soiling rule tuned on synthetic texture.
- **The hash chain detects ordinary edits;** it is not a signed audit log. The public demo has no login.

TRY IT

Open it, pick a sample, follow the numbers.

- `jgmzrkpa344jwixw7nbulgh2ju0mojcb.lambda-url.us-east-1.on.aws`
- `github.com/gmassello/afterimage`
- `gmassello.github.io/afterimage`
- `youtube.com/watch?v=zUFR96a33IM`